

Public Economics and Development

Week 12: Allocation of Authority Across Bureaucrats and Managers, and Corruption

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Why does the public sector personnel matter?

Take up huge roles: In principle and in data

- As in other sectors, they embody human capital of the state
- They determine provision of public services and implementation of policies
- Takes up nearly 20% of all employment and 35% of formal employment (World Bank)
- Responsible for the public sector that takes up approximately 45% of GDP (OECD)

However, the quality of public sector workers varies across countries

- 20-35% of front-line workers (teachers, health care) absent in developing countries
- Lack institutional structures to incentivize, monitor existing workers
 - ▶ Poor structure → Absenteeism and inefficiencies → Bad public services (repeat)
- Finding motivated and capable individuals to work is difficult
 - ▶ Lower pay, more rigid mobility than private sector
- Bureaucratic quality is also clustered with development

Public sector takes up huge share of employment globally

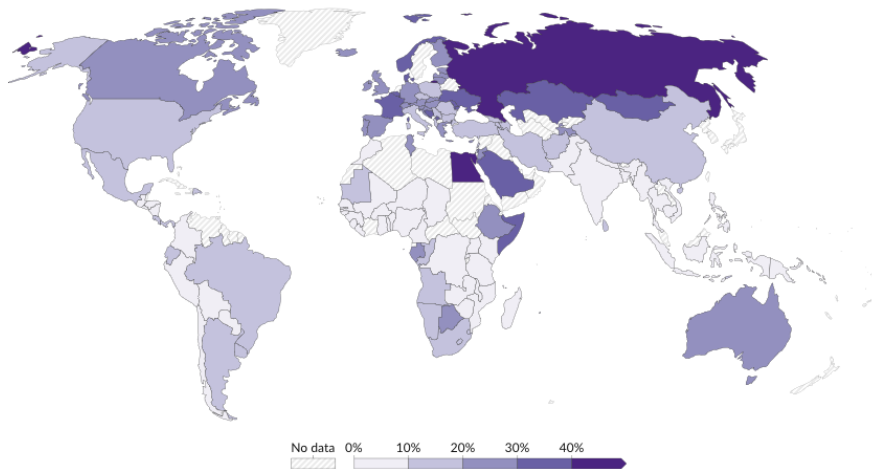
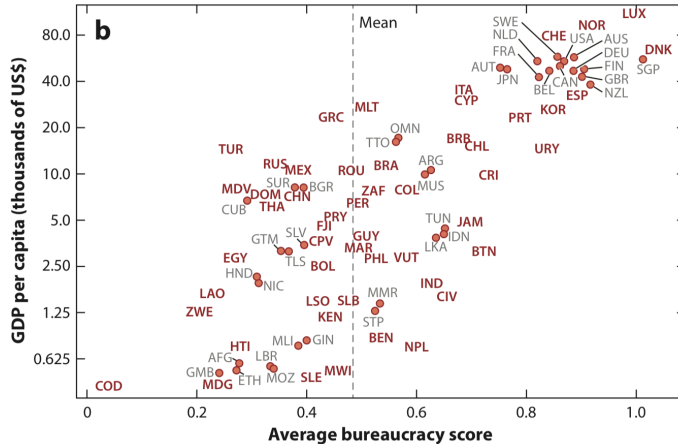


Figure: Share of public sector employment in total employment, 2019

Source: Our World in Data ([link](#)), reprocessed from World Bank and ILO data

Clustering of development and bureaucratic quality



Source: Besley, Burgess, Khan, Xu (2022) "Bureaucracy and Development", *Annual Review of Economics*, 14, 397-424

Corruption: A Global Development Challenge

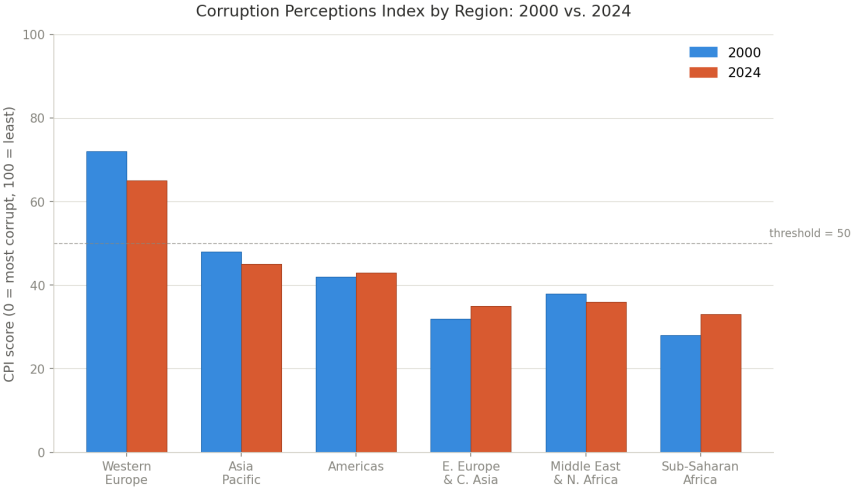
Corruption is rampant — especially in developing countries

- Over 2/3 of countries score below 50 on the Corruption Perceptions Index (CPI)
 - ▶ CPI ranges from 0 (highly corrupt) to 100 (very clean)
 - ▶ Global average CPI score stuck at 43 for over a decade — little progress
 - ▶ 148 out of 180 countries have stagnated or worsened since 2012
- Undermines public service delivery, economic growth, and trust in government
- Diverts public resources away from those who need them most

Yet research has lagged policy

- Hidden by definition — actors actively invest resources to conceal it
- Often associated with nonstate actors who capture state apparatus and media
 - ▶ Thus, difficult to get accurate estimates
 - ▶ Reliance on corruption perception as measurement, but it can be very subjective
- Theoretically difficult to model: Manifestation can vary in form across environments

Corruption Perceptions Index: Little Progress Over Two Decades



Source: Transparency International. Pre-2012 scores rescaled for comparability.

Source: Transparency International. CPI scale: 0 (highly corrupt) to 100 (very clean).

Roadmap for today

1. Overview setup of the topic

2. Allocating authority across bureaucrats and politicians

2.1 Aghion, Tirole (1997): Formal and Real Authority in Organizations

2.2 Bandiera, Best, Khan, Prat (2021): The Allocation of Authority in Organizations: A Field Experiment with Bureaucrats

3. Analysis of Corruption

3.1 Banerjee, Mullainathan, Hanna (2012): Corruption

3.2 Dal Bó, Dal Bó, and Di Tella (2006): Plata o Plomo? Bribe and Punishment in a Theory of Political Influence

3.3 Xu (2018): The Costs of Patronage: Evidence from the British Empire

Aghion, Tirole (1997):
Formal and Real Authority in Organizations

Q: What decides the allocation of authorities within organizations?

What do we mean by authority

- Formal authority: Legal right to select actions (who owns the decision, final say)
- Real authority: Effective control over decisions (who shapes the agenda)
- It is commonplace for these authorities to be held by different individuals
 - ▶ Those who hold formal authority may delegate to better-informed agents
 - ▶ Shareholders/Presidents (formal) vs Board of directors/Executive branch (real)
- Optimal decision-making structure when not everyone is equally informed?

This paper discusses optimal allocation and separation of different types of authorities

- A principal delegates a task (project selection) to an agent
- Both independently collect information about potential projects (at a cost)
- The agent recommends a project; the principal approves or vetoes
- Key question: When should the principal **retain** or **delegate** formal authority?

Model setup

Principal-agent can implement 0 or 1 among $n \geq 3$ projects

- Project k has profit B_k to principal and private benefit b_k to agent
- Principal's favorite project: B to the principal and βb to the agent ($\beta \in (0, 1]$)
- Agent's favorite project: b to the agent and αB to the principal ($\alpha \in (0, 1]$)
- Payoffs of the project initially unknown (perceived as identical)

Utility and cost to principal and agent

- Principal pays wage w to agent: Thus has utility $B_k - w$
- Agent has income w and private payoff b_k : Thus utility $u(w) + b_k$
- Gathering information is costly (even if effort is exerted may not learn anything at all)
 - ▶ Agent: Spend $g_A(e)$ to learn payoffs of all projects with probability e
 - ▶ Principal: Spend $g_P(E)$ to learn payoffs of all projects with probability E

Model setup

Timing of the decision structure

- Contract: Specifies wage and allocation of formal authority
- Principal and agent simultaneously choose information acquisition efforts (E and e)
- Agent observes own information and recommends a project (or stays silent)
- Principal observes own information and approves or vetoes the recommendation
- If no project is approved, status quo payoff of 0 is realized

Utility (from project selection) depending on who has formal authority

- Principal has formal authority (integration)

$$u_P = EB + (1 - E)e\alpha B - g_P(E) \quad u_A = E\beta b + (1 - E)eb - g_A(e)$$

- Agent has formal authority (delegation)

$$u_P^d = e\alpha B + (1 - e)EB - g_P(E) \quad u_A^d = eb + (1 - e)E\beta b - g_A(e)$$

Key trade-off: Principal's control vs. Agent's initiative

Efforts of principal and agents are *strategic substitutes*: Principal's $E \uparrow \rightarrow$ Agent's $e \downarrow$

- The first order conditions in the integration case is

$$P : (1 - e\alpha)B = g'_P(E) \quad A : (1 - E)b = g'_A(e)$$

- FOC under delegation

$$P : (1 - e)B = g'_P(E) \quad A : (1 - E\beta)b = g'_A(e)$$

- For instance, suppose that MC of searching increases in integrated case ($g'_p \uparrow$)
 - ▶ Probability of being informed about payoffs decreases $E \downarrow \rightarrow$ Principal loses control
 - ▶ But it can encourage agent to take initiative ($e \uparrow$)
- You can argue similarly for delegation case (A loses initiative $\rightarrow$ P takes control)
- Effort to collect information increases with the following factors..
 - ▶ Principal: Lower congruence (α), higher stakes (B), lower agent effort (e)
 - ▶ Agent: Lower congruence (β), higher private gains (b), lower principal interference (E)

Optimal formal authority: When should principal delegate?

Comparing equilibrium under integration $\{E, e\}$ and delegation $\{E^d, e^d\}$

- $e^d > e$: Agent exerts **greater initiative** under delegation
- $E^d < E$: Principal exerts **less effort** — loses both formal and real authority
- Agent more likely to find preferred project $\Rightarrow$ higher payoff $b_k \Rightarrow$ accepts lower wage

Principal prefers to delegate over integration when initiative gains outweigh loss of control

- High congruence ($\alpha \uparrow$): Agent's preferred project less costly to principal
- High private gains ($b \uparrow$): Agent more motivated to acquire information
- High cost of principal's information acquisition ($g'_p \uparrow$): Retaining control is costly

Principal is fine with losing control if agents can find and implement good projects

- Especially when principal is overloaded, there are many superiors, and matter is urgent

Span of control: Optimal # of agents and degree of overload?

Suppose that

- Principal retains formal authority
- Hires m identical agents, and monitoring them is costlier

Adding an extra agent has two opposing effects:

- Overload cost: Extra agent costs more in attention than revenue gained
- Initiative effect: $e \uparrow \Rightarrow$ agents find better projects $\Rightarrow$ principal's revenue increases

At the optimum, overload cost exactly offsets initiative effect

- Principal overhires beyond the point where marginal profit is zero
- The extra cost is made up for by gains from increased initiative
- Busy principal cannot micromanage $\rightarrow$ Overload as commitment not to stifle initiatives

Key takeaways and discussions

Other findings

- Participation view: delegate decisions that matter more to agent than principal
- Higher wages increase agent's real authority
 - ▶ Stronger incentives → agent more likely to make recommendations
 - ▶ High wage reduce principal's net profit ⇒ Monitor less ⇒ less likely to overrule agent
- Integration may jeopardize communication
 - ▶ When congruence is low, agent more likely to withhold information

Open questions

- How does optimal authority allocation change with multilayered hierarchies?
 - ▶ Many middle managers or many levels of subnational government
- Authority or monetary incentives: what attracts and motivates agents?
 - ▶ Agents whose decisions are not overruled may be more motivated to stay
 - ▶ But others may value monetary incentives over decision rights

Bandiera, Best, Khan, Prat (2021):
The Allocation of Authority in Organizations: A
Field Experiment with Bureaucrats

Q: How does allocation of authority to bureaucrats affect outcomes?

Organization bring together people with different interests and skills

- Working towards a common goal challenging: Principal-agent problem and monitoring
- How to allocate decision rights at different layers of hierarchy?
- How to monitor and incentivize agents' behavior?
- Many work on effects of performance pay, few look at decision-making structure

This paper designs an experiment in Pakistan procurement sector that combines the two

- Strict rules and monitoring vs. simplification and more autonomy?
- Procurement officers (PO, make purchases) and Monitors (approve purchase)
 - ▶ Monitors: Approve purchases quickly vs Delay purchase until end of fiscal year
- Experiment: Control + Autonomy + Performance pay + Both
- Different policy instrument works in different structures

Procurement structure in Pakistan and theoretical framework

Procurement structure

- Experiment took place in 26 / 36 districts in Punjab (80% of population)
- Each office has 1 PO with allocated budgets
- POs submit purchase for pre-audit approval to a monitor in each district
 - ▶ This monitor can ask for extra paperwork (withhold) or approve the purchase

Conceptual framework: Expected effects of autonomy vs incentives by type of monitors

- Aligned PO and monitors behave in the interest of citizens (low price, little delays)
- Misaligned PO raises price (bribes, little effort to find low-price)
- Misaligned monitor delays purchase (exert power on PO)
- Autonomy: Price↓ if PO more aligned then monitor (opposite if PO more misaligned)
- Incentives: Motivate PO, but dampened with misaligned monitors (higher costs)

Experimental design

Measuring bureaucratic performance with value for money (with online platform)

- Price conditional on quantity, nature of purchased goods, and net of transaction costs
 - ▶ Items: Consumables such as papers, toners, pen, banners, light bulbs etc
 - ▶ Price is adjusted for delivery speed, cost of delivery
- Others: Expected price, good-bad purchase categorization(based on attributes), ML

Experimental design

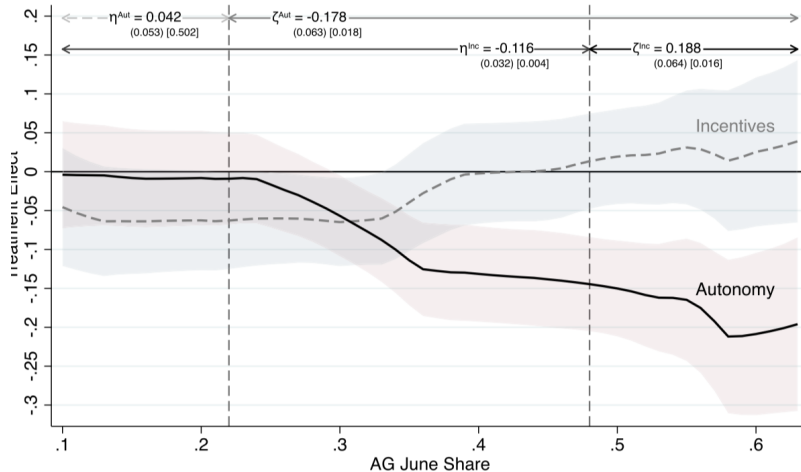
- Autonomy: Remove the role of monitors in purchasing decisions
- Incentive: Range of bonus for improving value for money ($\times 2$ - $\times 1/2$ of monthly wage)
- Another group that receives both
- Stratified by district-department level

$$y_{igto} = \alpha + \sum_{k=1}^3 \eta_k \text{Treatment}_o^k + \rho_g q_{igto} + \delta_s + \gamma_g + e_{igto}$$

Overall, autonomy → lower prices paid but not incentives

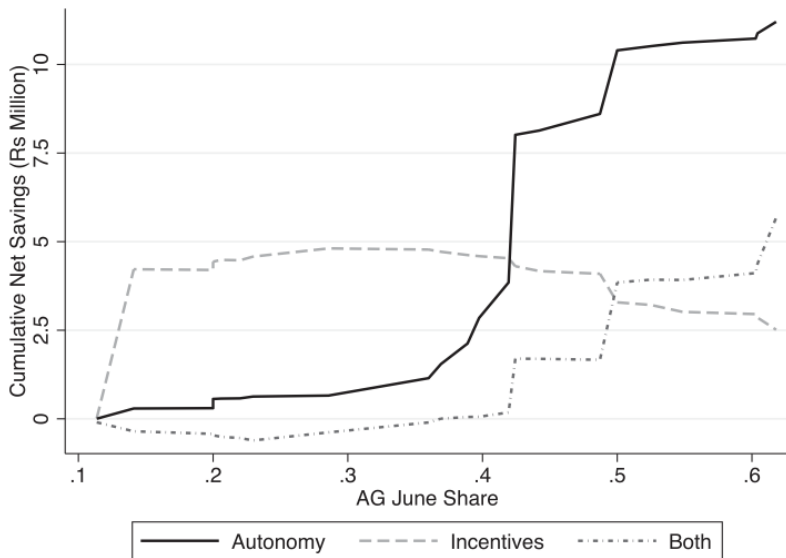
	Variety			Unit price				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Autonomy	0.016 (0.030) [.646]	0.010 (0.023) [.705]	-0.000 (0.009) [.819]	-0.085 (0.038) [.046]	-0.086 (0.032) [.018]	-0.080 (0.031) [.023]	-0.082 (0.034) [.030]	-0.085 (0.038) [.047]
Incentives	0.006 (0.030) [.846]	0.025 (0.023) [.325]	0.005 (0.009) [.382]	-0.016 (0.038) [.723]	-0.026 (0.030) [.476]	-0.022 (0.033) [.571]	-0.020 (0.034) [.625]	-0.017 (0.038) [.717]
Both	0.037 (0.030) [.265]	0.059 (0.023) [.021]	0.006 (0.009) [.484]	-0.070 (0.041) [.130]	-0.083 (0.032) [.025]	-0.072 (0.033) [.053]	-0.086 (0.039) [.043]	-0.070 (0.041) [.127]
Item variety measure	Scalar	Coarse	ML	None	Attribs	Scalar	Coarse	ML
$p(\text{all} = 0)$	.660	.080	.777	.168	.054	.093	.087	.165
$p(\text{autonomy} = \text{incentives})$	.749	.537	.447	.146	.077	.119	.119	.152
$p(\text{autonomy} = \text{both})$	.461	.031	.566	.741	.927	.807	.932	.747
$p(\text{incentives} = \text{both})$	.302	.144	.953	.262	.133	.227	.136	.257
Observations	11,771	11,771	11,771	11,771	11,771	11,771	11,771	11,771

Effects differ by type of monitors



- Better monitors (low delays, AG June share ↓): Incentive lower price (not autonomy)
- Worse monitors (more delays, AG June share ↑): Autonomy lower price (not incentives)

Treatment → large savings, but heterogenous by monitor types



Takeaways and remaining questions

Incentives and autonomy work in different situations

- Incentives to low-level agents mean little if their supervisors are misaligned
- In such case, giving low-level agents more autonomy improves performance
- If supervisors are aligned, incentives improve effort of agents
- On the other hand, autonomy does not change much from status quo in this case

Remaining questions on the organizational Economics

- Can stringent rules, not autonomy, be an optimal choice?
 - ▶ Could still be worthwhile to prevent misaligned agents imposing high costs
- How does autonomy affect *who becomes a bureaucrat*?
 - ▶ How to bring in more aligned agents (make efficient decisions for public) than misaligned agents (extract personal benefits with autonomy)?

Banerjee, Mullainathan, Hanna (2012):
Corruption

Q: What are the causes and consequences of corruption?

Despite rampant corruption, research has lagged reality

- Corruption is empirically difficult to measure
- Theoretical analysis focused exclusively on moral hazard in organizations
 - ▶ Emphasis on corruption as a criminal action requiring monitoring and strict penalties
 - ▶ In practice, corruption can manifest in many different forms across different environment
- Need to understand corruption from a different standpoint

This paper provides a novel framework for analyzing corruption in public organizations

- Corruption as result of tasks, private incentives, and organizational environment
- Highlight how discretion (vs. rules) affects corruption in different environments
- Also study other distortions that occur with corruption
 - ▶ Red-tape, inefficiencies in service provision

What is corruption

Rule-breaking by a bureaucrat (or elected official) for **private gain**

- Overt monetary bribe to bend a rule and provide service to unintended recipient
- Nepotism as opposed to competitive hiring / selection
- Not showing up to work ('steal' time)
- These are defined by rules specific to a context

How this definition differs from other approaches

- Emphasis on breaking the rules (previous definitions emphasize personal gain)
 - ▶ A boss *with final say* hiring his nephew: Not corruption if no other rules broken
 - ▶ Previous definitions would count this as corruption (personal gain)
- Advantages of this definition:
 - ▶ Avoids subjective ethical judgements — rules define what is corrupt
 - ▶ Encompasses non-monetary gains (nepotism, time theft)
 - ▶ Rule-breaking is easier to measure than bribes

Model setup

A bureaucrat assigns a limited number of slots to a population of size N

- Individuals differ in social value, private value, and ability to pay
- Individuals announce their types (private value is hidden)

Bureaucrat announces mechanism specifying testing, prices, and assignment probabilities

- Testing: screen individual's types at a cost to both bureaucrat and individual
- Assignment: probability of receiving a slot given test outcome
- Prices: what individuals pay for a slot

Government sets rules that constrain the mechanism (and define what is corrupt)

- If rules don't pin down a unique mechanism $\Rightarrow$ bureaucrat has **discretion**
- Various dimensions of malfeasance beyond corruption
 - ▶ Overcharging price (bribe), excessive testing (red-tape), no testing (shirking)
 - ▶ Misallocation (allocative inefficiency): slots assigned to wrong types or left unassigned

The task approach to corruption

Corruption arises from the nature of the task itself

- Same bureaucrat, same incentives — different tasks generate different corruption
- Can we eliminate corruption by changing the rules without affecting anything else?

Key tradeoff between setting strict rules and allowing discretion

- Strict rules limit corruption but generates red tape and misallocation
 - ▶ Could ensure that public services are allocated based on needs (welfare)
 - ▶ Strict rules bind honest bureaucrats but corrupt ones find ways around them
- Full discretion reduces red tape but opens door to corruption
 - ▶ Italian procurement: Fixed budget with full discretion reduced waste and corruption
 - ▶ Hospital bed allocation: May lead to allocation to who can pay, not need-based

Optimal response: redesign the task (rules & discretion), not just increase punishment

- Goal is to improve public service provision (punishment is one of the tools)

Manifestations of corruption

The form corruption takes depends on rules and gap between social and private values

- Bribes: Bureaucrat has pricing discretion $\Rightarrow$ overcharges those who can pay
 - ▶ Arises when private value exceeds social value – bureaucrat exploits the gap
- Red tape: Bureaucrat cannot price freely $\Rightarrow$ imposes excessive testing costs
- Misallocation: Bureaucrat has assignment discretion $\Rightarrow$ slots go to those who can pay
 - ▶ Arises when social value is high but ability to pay is low (e.g. poor patient, small farmer)

These distortions arise simultaneously and reducing one dimension may worsen others

- Restrict pricing $\Rightarrow$ Bribes fall but red tape rises
- Restrict testing $\Rightarrow$ Red tape falls but misallocation rises
- Restrict assignment $\Rightarrow$ Misallocation falls but bribes rise (value in bending rules)
- Optimal policy: redesign the task so that corruption causes the least harm

Measuring corruption: Empirical strategies used in the literature

Perception-based measures (old approach): surveys of participants

- Problems: subjective, culturally biased, hard to compare across settings

Objective measures (new approach): direct measurement or administrative data

- Price comparisons (procurement): Price discrepancies for identical goods
- Audit studies: Send trained auditors to measure bribes and red tape directly
- Absenteeism: Photograph or timestamp bureaucrat presence — captures time theft
- Leakage studies: Track whether public resources actually reach intended recipients

Key challenge: Measurement itself changes behavior (Hawthorne effect)

- Actors reduce illicit behavior during measurement or find new ways to obscure it

Policy Implications and Open Questions

What works against corruption?

- Monitoring / punishment: Effective but limited — bureaucrats adapt and hide behavior
- Task redesign: Change rules and discretion to minimize the most costly distortion
- Wages: Higher wages may reduce corruption by raising cost of job loss
- Competition: Multiple providers reduces bureaucrat's monopoly over slots
- Nonmonetary benefits: Attract intrinsically motivated, prosocial bureaucrats

Open questions

- How to recruit intrinsically motivated individuals to public sector?
 - ▶ Public sector competes with private sector
 - ▶ Higher wages *could* work, but may also attract undesirable individuals
- How does allowing discretion deter corruption in practice?
 - ▶ Could increase initiative and overrule corruptible middle managers
 - ▶ But also increase scope for corruption due to less interventions

Dal Bó, Dal Bó, and Di Tella (2006):
Plata o Plomo? Bribe and Punishment in a
Theory of Political Influence

Q: How do state capture affect quality of public officials

In weak states, politicians are subject to punishments by private entities and bribes

- Many private entities seek to influence politicians through bribes and coercion
- Punishments and bribes complement: Threat of punishment makes bribing easier
- Yet, there are things where models with only corruption cannot explain
 - ▶ Why give officials more discretion and immunity?

This paper updates framework of political influence to include bribes and coercion

- Investigates how possibility of coercion affects quality of public workers
- Explains why political immunities exist in contexts where private coercion is prevalent
- Explains how discretion actually makes public work less attractive
 - ▶ In model with bribes only, immunities are rarely justified (weaken monitoring mechanisms)
 - ▶ Also in the old model, discretion increases corruption and appeal of public work

Model setup: Decision to join public work

First stage: Citizens decide to enter public office or not

- Citizens endowed with ability $a \in [0, \infty)$ (a probability distribution)
- Alternative is to choose private sector (which gives wage a , equivalent to their ability)
- Return from public work: public sector wage (w) + behavior of pressure group
- Citizens with $a \leq w$ apply (and w types selected, assuming recruiters observe types)

Second stage: Pressure group has an opportunity to influence officials

- Public workers provide public good, whose quality positively correlates with ability
- Have discretion to allocate (π) towards pressure group, with cost c if caught
- With probability γ , officials decide to accept bribes or face punishments
 - ▶ $1 - \gamma$: Only punishment (no possibility to accept bribes) - when bribes are infeasible
- Pressure groups can either bribe (b) or punish (p) at costs $\beta\Phi(b)$ and $\rho\Psi(p)$

Analysis of payoffs

Decision on accepting (and handing out) bribes

- Officials facing pressure groups accept bribes (share γ) if

$$w + b - c \geq w - p$$

- Pressure groups sets b and p to maximize the following profits

$$\Pi(b, p) = \gamma(\pi - \beta\Phi(b)) - (1 - \gamma)\rho\Psi(p) \quad \text{s.t.} \quad b \geq c - p$$

- $\tilde{\pi}$: Minimal discretion required for pressure groups to influence officials
 - ▶ π that makes $\Pi(b^*, p^*) = 0$ (π that makes profits when groups optimize 0)
 - ▶ Inversely related to state capture: $\tilde{\pi} \uparrow \rightarrow$ Cost of bribing an official $\uparrow \rightarrow$ State capture $\downarrow$

If only bribes are available ($p = 0$ for all cases)

- Solve $\gamma(\pi - \beta\Psi(b))$ subject to $b \geq c \rightarrow$ Bribe just enough to cover c , ($b = c$)
- Official payoffs/skill is w , and $\tilde{\pi}^0 = \beta\Phi(c)$
- Drop in bribe costs ($\beta \downarrow$) $\rightarrow$ Higher state capture ($\tilde{\pi} \downarrow$), no effect on quality of officials

What happens when both bribes and coercion are possible?

Problem when both b and p are active

- Pay bribes just enough to cover $b = c - p$ (Bribes become cheaper)
- The shortfall is made up with nonzero punishment p^* , such that $b + p^* = c$
- $\tilde{\pi}^P$ satisfies $\frac{1-\gamma}{\gamma}\rho\Psi(p^*) + \beta\Phi(c - p^*) \rightarrow$ Any discretion π larger than this, then influence
- Resulting equilibrium payoff of official if pressure groups are active
 - ▶ If official refuses bribe, payoff is $w - p^*$ (lower due to nonzero punishments)
 - ▶ If bribes are accepted, payoff is $w + b - c = w + c - p^* - c = w - p^*$
- When pressure groups are inactive: payoff is $w > w - p^*$
- Thus, the quality of bureaucrats fall in this scenario
- Furthermore, $\tilde{\pi}^P < \tilde{\pi}^0 \rightarrow$ When threats are usable, more state capture!

Negative effects amplified with cheaper ways to influence officials

Exogenous changes in cost of influence (Suppose that β and ρ drops)

- Recall that $\tilde{\pi}^P = \frac{1-\gamma}{\gamma}\rho\Psi(p^*) + \beta\Phi(c - p^*)$ and $\Pi(b, p) = \gamma(\pi - \beta\Phi(b)) - (1 - \gamma)\rho\Psi(p)$
- Fall in any one of β and ρ raises profits and decreases $\tilde{\pi}^P$
 - ▶ State capture will be more rampant
 - ▶ Existing pressure groups ramp up activity + new pressure groups enter
- If primarily driven by $\rho \downarrow$, then payoff of officials (and quality) definitely falls
 - ▶ Usage of coercion increases in any case
- If primarily driven by $\beta \downarrow$, change to pay off ambiguous (but definitely not increase)
 - ▶ Existing groups shift away from coercion → Minimal effect on wages
 - ▶ New entrants that use both coercion and bribes → lower payoffs

Implication on discretions

How do discretion affect quality of politicians

- If discretion $\pi < \tilde{\pi}^P$, less incentives for pressure groups to influence officials
- Higher discretion could lead to more waste (more corrupt decision)
- As more groups with coercion decrease payoffs, quality of officials worsened

Immunity of officials (in principle)

- Can insulate honest officials from outside pressures, mitigate false accusations
- But can mask corrupt officials
- In setting with rampant coercion from pressure groups
 - ▶ There are greater effect of protection to officials with immunity
 - ▶ This can overweigh bad effects of masking corrupt officials
- When legal system is effective, masking effects outweigh pressure effects

Takeaways and discussions

Threat of coercion imposes negative effects on governance

- Reduces payoff for officials → lower quality of officials
- More influence from private pressure groups → Increased state capture
- Augments effects of bribes: Cheaper to bribe officials

Remaining questions

- Ultimately, need to improve legal capacity
 - ▶ To prevent pressure groups and detect corruption
 - ▶ But transitioning to better legal system is tough
- Incentives of bureaucrats: How to attract incorruptible moral officials?
 - ▶ Under what conditions can governments bring them in (what sort of incentives?)
 - ▶ Topic of Dal Bó et al (2013)

Xu (2018):
The Costs of Patronage: Evidence from the
British Empire

Q: How does patronage affect allocation of positions & performance?

How personnel are promoted and incentivized determines organizational performance

- Past: patronage widely used to appoint public sector positions
 - ▶ Discretionary appointments based on network and ties to the ruler
- In theory, discretion can align incentives between principal and agents
- However, it can worsen performance if favoritism disincentivizes capable subordinates

This paper uses historical data to identify how patronage affects performance

- Colonial governors of British empire: 1854-1966 across 70 colonies
- Connection determined by genealogy and biological data
- Within-governor variation: Change in secretary of state → Change in connections
- Across-time variation: Discretion (1854-1930) vs. Independent appointment (1930 -)
- Test who gets assigned to richer colonies and how tax returns are affected

Organization of British colonies

Assignment of colonial governors

- Handled by Colonial Office under Secretary of State for the Colonies in London
- Secretary of State in charge for around 3 years
- 1930 Reform replace Secretary of State's discretion with open recruitment

Role of the governors

- Fixed 6-year term
- Responsible for public finance, legislation, and personnel appointments
- Discretion over executive and finances: Widely different policies across colonies
- Direct monitoring impossible due to distance

Data and empirical strategy

Digitalized (by Guo Xu himself!) archival data

- Sources: Colonial Blue Books, Colonial Office List, Who's Who, The Peerage
- Postings (and their wages), background of governors, genealogy
 - ▶ Connection determined by ancestry, membership at aristocratic clubs, and schools
- Colonial statistics (revenue, expenditure, demographics etc)

Empirical strategy leverages differences in connectedness across different periods

$$y_{it} = \beta c_{it} + \theta_i + \gamma X_{it} + \tau_t + e_{it}$$

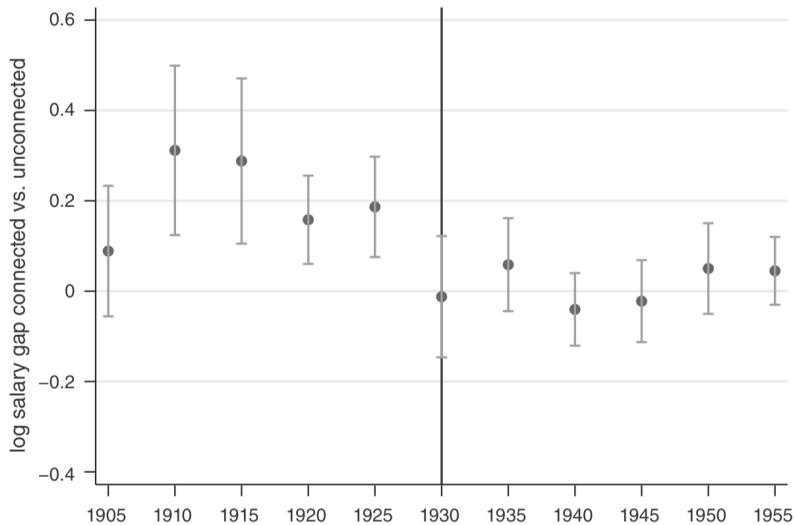
$$y_{it} = \beta_0 c_{it} + \beta_1 c_{it} \times I[t \geq 1930] + \theta_i + \gamma X_{it} + \tau_t + e_{it}$$

- Connection c_{it} used with various criterion
- c_{it} is predetermined and driven by secretary of state changes
 - ▶ No reverse causality: Unlikely to attend Oxbridge to be connected to current governor
 - ▶ Secretary of state turnover generates common shocks to all serving governors

Connected governor salaries are higher!

	log governor salary in GBP					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean of dep. var.	7.929	7.929	7.929	7.929	7.929	7.929
No. colonies served	0.221 (0.035)	0.222 (0.035)	0.223 (0.035)	0.222 (0.035)	0.224 (0.035)	0.223 (0.035)
Shared ancestors	0.103 (0.047)				0.093 (0.046)	
Both aristocrats		0.215 (0.124)			0.176 (0.121)	
Both Eton			0.133 (0.077)		0.118 (0.081)	
Both Oxbridge				0.072 (0.047)	0.073 (0.045)	
Connected						0.097 (0.036)
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Governor FEs	Yes	Yes	Yes	Yes	Yes	Yes
Spell length FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,510	3,510	3,510	3,510	3,510	3,510

Salary gap washed away post-1930



Patronage worsens performance, which is offset following reforms

	Colony-level				Public expenditure			
	Public review		Public expenditures		Overall		Tax	Works
	Overall		Trade	Internal	(5)	(6)	(7)	(8)
	(1)	(2)	(3)	(4)				
<i>Panel A. Revenue</i>								
Mean of dep. var.	12.31	12.31	11.47	11.59	12.33	12.37	9.015	10.32
Connected	-0.040 (0.017)	-0.055 (0.021)	-0.053 (0.026)	-0.043 (0.032)	-0.029 (0.019)	-0.042 (0.023)	-0.089 (0.053)	-0.107 (0.062)
Connected × reform dummy		0.061 (0.033)				0.053 (0.034)		
Connected + connected × reform dummy	-	0.005 (0.026)	-	-	-	0.010 (0.025)	-	-
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Governor-Colony FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Spell length FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time-varying controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,510	3,510	2,670	2,657	3,510	3,510	1,742	2,588

Takeaways and discussions

What the findings say about organizational performance

- Connection → Positions that yield higher wages
- Lower revenue collection: Due to higher tax exemptions, high reliance on customs
- Reforms washed salary gaps and shortfalls in performance

Discussions

- Favoritism can lead to worse organizational performance, even in public institutions
- But how does it affect recruitment into the pool of governor candidates?
 - ▶ If people know connections matter, talented individuals with no connection discouraged
 - ▶ Leaving pool of candidates with less talented individuals
 - ▶ Ways to look for 'outsiders' with talent?

Remaining questions

How to attract talented individuals to the public sector?

- In a lot of places, they have lower pay
- But public sector offers different perks: Stability, social respect, intrinsic motivation
- How can these individuals be recruited?
 - ▶ Does financial incentives work?
 - ▶ Are these individuals different from the general public?

How to design a healthy public organizations?

- Healthy balance of incentives and authority that prevents misallocation and corruption
- Monitoring and incentives are still often used
- But are there other aspects of public organizations that can be used to improve performance?
 - ▶ Career progression: Initial and subsequent posting to different positions
 - ▶ Middle managers: Can they bridge the gap between agents and the principal (the people)